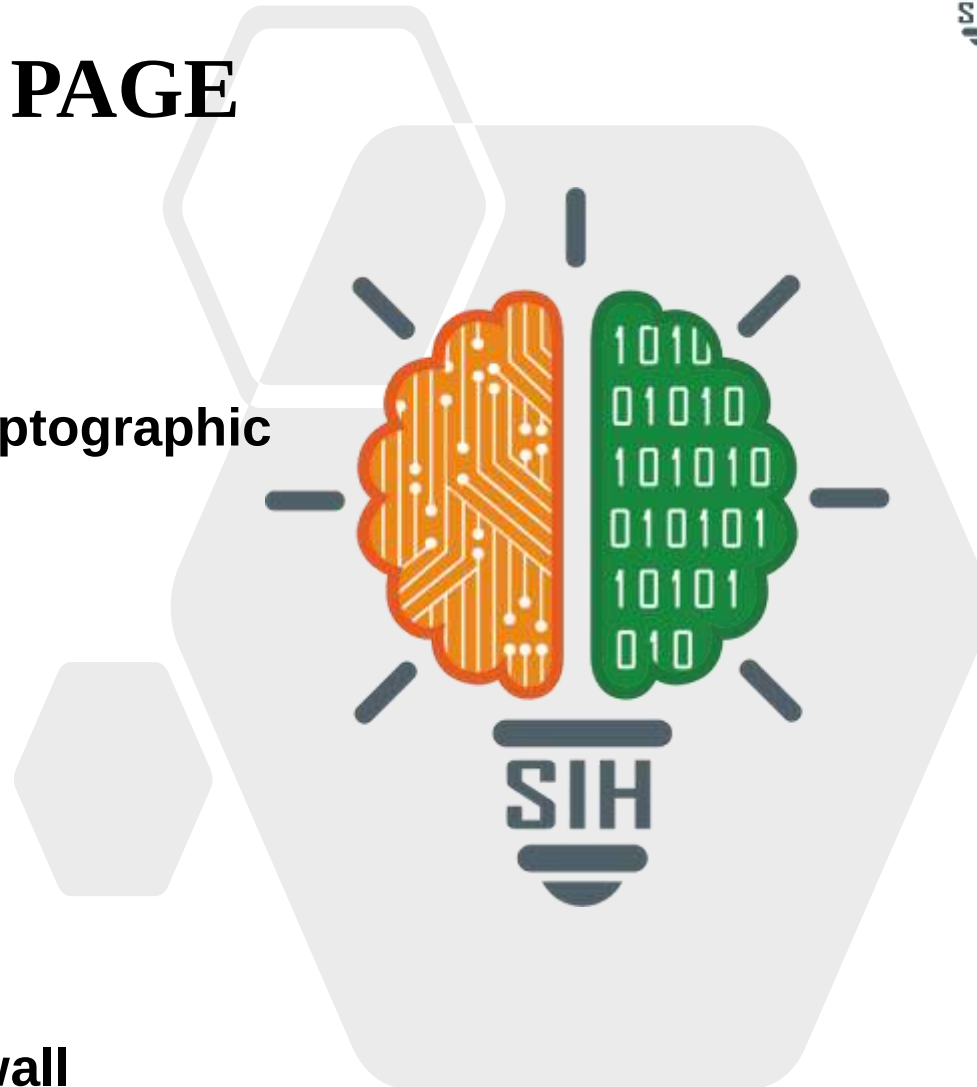


TITLE PAGE

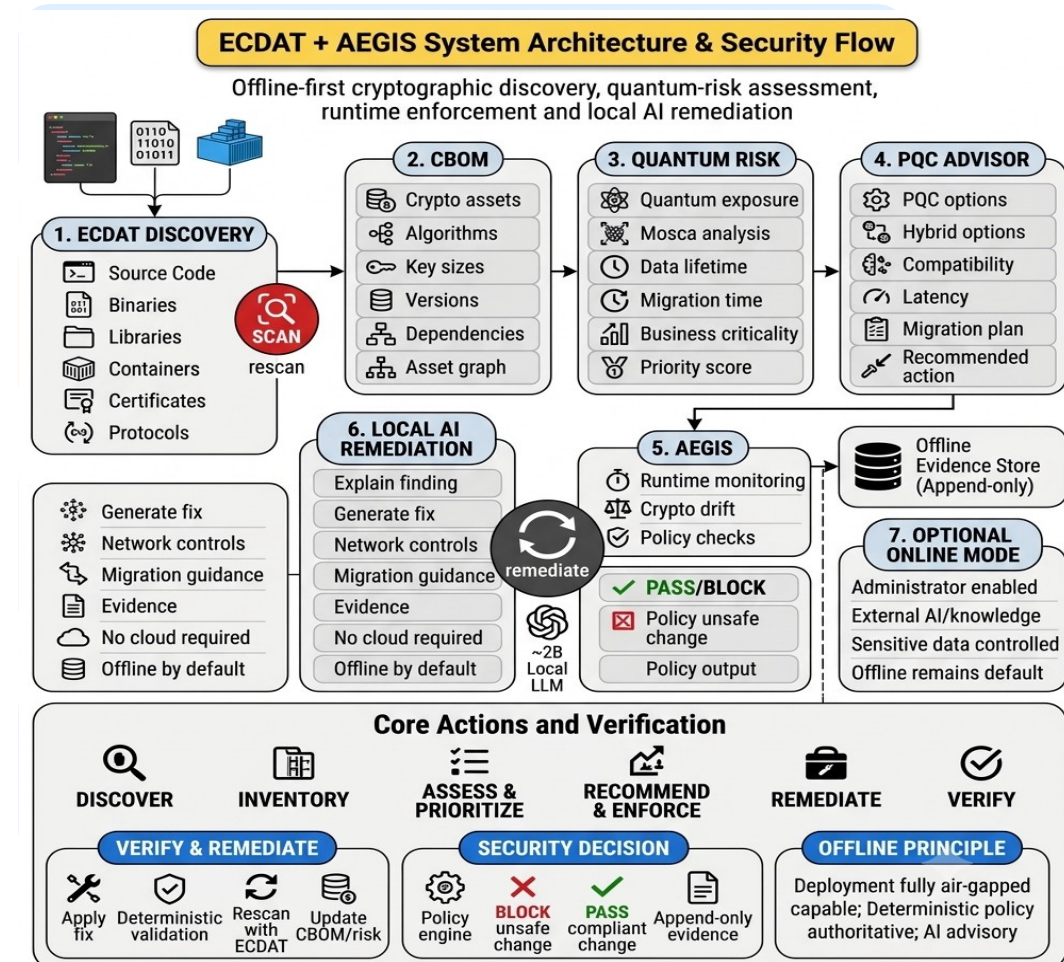
- Problem Statement ID – SIH26164
- Problem Statement Title - Enterprise Cryptographic
Discovery & Analysis Tool (ECDAT)
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID-
- Team Name (Registered on portal): Firewall



Proposed Solution:

- **Enterprise Cryptographic Discovery:** Automatically discovers algorithms, keys, certificates, protocols and crypto dependencies across source code, binaries, libraries, containers and infrastructure.
- **Cryptographic Bill of Materials (CBOM):** Builds a centralized inventory of crypto assets, including algorithms, versions, key sizes, usage patterns and dependency graph.
- **Quantum Risk Analyzer:** Evaluates assets against quantum threats using Mosca's approach to flag systems that require urgent cryptographic migration.
- **PQC Migration Advisor :** Recommends suitable PQC and hybrid cryptographic options based on security risk, system constraints and migration priority.
- **AEGIS Runtime Security:** Monitors live cryptographic activity, detects drift and unsafe changes, and enforces security policies in real time.
- **Offline AI Remediation:** Uses a locally deployed ~2B parameter LLM to explain findings and generate remediation guidance without sending source code to the cloud.
- **Verify & Enforce:** AEGIS validates proposed changes (PASS/BLOCK) and ECDAT rescans systems to confirm that overall cryptographic posture has improved

System Architecture :



A modular C++20 security engine linking cryptographic discovery, quantum-risk assessment, migration planning, and AEGIS runtime enforcement.

TECHNOLOGIES USED

**C++20**

Security Engine

**CMake**

Cross-Platform Build

**CLI11**

Command Line Interface

**nlohmann/json + yaml-cpp**

Asset & taxonomy data

**SQLite3**

Local audits & scan history

**AEGIS + Docker**

Runtime observation

**spdlog + fmt**

Structured logging

**Tree-sitter + Python**

AST discovery

**OpenSSL / X.509**

PEM certificate discovery

**TLS Scanner**

Weak protocol detection

**Risk + PQC Engine**

Quantum risk, readiness & migration

**Local AI + Deterministic Gate**

Redacted remediation

METHODOLOGY



DISCOVER & INVENTORY

Baseline repository state → Source / Certificate / TLS discovery → Unified CryptoAsset / CBOM



CLASSIFY & ASSESS

Taxonomy → Weakness / Exposure → Quantum Risk → PQC Readiness



PRIORITIZE & RECOMMEND

Mosca prioritization → Migration Readiness → PQC / Hybrid Recommendation



OBSERVE, DETECT & ENFORCE

AEGIS Runtime Observer → Crypto Drift → Policy Evaluation → Pass / Block

A practical, modular architecture with offline-first deployment, explainable analysis, and controlled extension toward AEGIS enforcement.

1 TECHNICAL FEASIBILITY

- Native C++20 implementation with CMake
- Tree-sitter for crypto discovery in Python code
- OpenSSL for X.509 & TLS configuration analysis
- SQLite for results; JSON / CSV / PDF reports

2 DEPLOYMENT FEASIBILITY

- Pre-built, self-contained packages
- No development dependencies for users
- Offline-first operation for secure environments
- Portable and reproducible workflow

3 OPERATIONAL VIABILITY

- End-to-end: Discover → Classify → Risk/PQC → Report
- Consistent security classification via taxonomy
- Explainable insights: What, Where, Why, Action
- Risk scoring to prioritize remediation

4 RISKS & MITIGATION

- Python-focused discovery → extend to more languages
- TLS analysis uses configuration patterns, not live handshake
- Expand algorithm taxonomy over time
- Cross-platform validation + automated regression tests

FEASIBILITY CHECK Buildable with existing components • Deployable locally • Explainable output • Risks have clear mitigation paths

IMPACT AND BENEFITS

IMPACT:

- **Continuous Protection:** Moves beyond finding cryptography to continuously understanding and protecting cryptographic posture.
- **Quantum Ready Migration:** Prioritizes quantum vulnerable cryptography using lifetime, criticality, exposure and migration effort.
- **Prevents Crypto Drift:** Detects weaker algorithms, protocol downgrades and cryptographic regressions against a trusted baseline.
- **Actionable Migration:** Connects cryptographic risks to suitable PQC/hybrid migration paths.
- **Enforces block before completion:** the core tool operates locally without a server, cloud service, internet connection, external API or Python runtime, hence blocking unsafe cryptographic changes.

**DISCOVER**Identify
dependencies**ASSESS**Evaluate quantum
risk**ENFORCE**Apply security
policy**REMEDiate**Execute guided
risk**VERIFY**Validate and
Rescan

KEY BENEFITS:

**Unified cryptographic visibility —**
Cryptographic assets, dependencies and relationships consolidated into a unified Crypto Asset / CBOM view..**Quantum Risk Prioritization —**
Combines exposure, lifetime, criticality and migration effort to determine what needs attention first.**Runtime Enforcement + Drift —**
AEGIS turns cryptographic policy into an enforcement point and detects deviations from the approved baseline.**Deterministic Verification —**
AI may assist remediation, but deterministic security policy remains authoritative. Fixes must pass validation and an ECDAT rescan.

Capability	IBM Quantum Safe Explorer	Qinsight Atlas	Comcast CARAF	pqAgility	ECDAT + AEGIS
Crypto discovery	✓	✓	✓	✓	✓
CBOM generation	✓	✓	Limited	✓	✓
Source-code scanning	-	✓	-	Limited	✓
Binary / container analysis	-	-	✓	-	✓
Quantum-risk assessment	✓	✓	✓	✓	✓
Migration prioritization	✓	✓	Limited	✓	✓
PQC recommendations	✓	✓	✓	✓	✓
Runtime crypto monitoring	-	-	-	-	✓
Crypto-drift detection	-	-	-	-	✓
Runtime enforcement	-	-	-	-	✓
Block unsafe crypto changes	-	-	-	-	✓
Local LLM remediation	-	-	-	-	✓
Fully offline AI	-	-	-	-	✓
Optional online AI	-	-	-	-	✓
Air-gapped deployment	-	-	-	-	✓
AI-agent protection	-	-	-	-	✓
Deterministic security gate	-	-	-	-	✓
Automated remediation loop	-	-	-	-	✓
Security evidence / audit trail	✓	✓	✓	Limited	✓

REFERENCES

1

NIST — Migration to Post-Quantum Cryptography / NCCoE
Cryptographic discovery, inventory, risk assessment, migration planning

2

NIST — Post-Quantum Cryptography
PQC standards: **FIPS 203 (ML-KEM)**, **FIPS 204 (ML-DSA)**, **FIPS 205 (SLH-DSA)**

3

NIST CSWP 39upd1 —
Considerations for Achieving Crypto Agility
Crypto-agility and ability to replace/adapt cryptographic algorithms

4

NIST — PQC Migration FAQs
Cryptographic inventory, CBOM relevance, discovery and migration phases

5

NTRO SIH Problem Statement 26164
The actual requirements of your SIH problem: discovery, catalogue, quantum risk, Mosca analysis, PQC/hybrid recommendations, CBOM and GUI

SIH Internal Shortlisted Idea -

<https://github.com/Roshan-Mallick/aegis-preflight-cli>

This idea was shortlisted during the SIH Internal Hackathon and has now been integrated into our current ECDAT solution as a runtime security and enforcement layer.